25901197

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECCSC202 - VISUALIZATION TOOLS FOR DATA ANALYTICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Scheme Of Evaluation

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. State the purpose of Google Data Studio as a data visualization tool. [K] / [CO1]

- **Converts raw data into interactive dashboards – 1 mark**
- **Helps analyze and share insights visually – 1 mark**

2. Difference between Bullet Chart and Time Series Chart. [U] / [CO2]

Bullet Chart – 1 mark

- Compares performance with target

Time Series Chart – 1 mark

- Shows trends over time

3. What is meant by collaborative analytics in Power BI? [U] / [CO3]

Definition – 2 marks

- Multiple users work on reports/dashboards – 1 mark
- Enables sharing and real-time collaboration – 1 mark

4. State the data aspects addressed by diagnostic analytics. [U] / [CO1]

Any two aspects – 1 mark each

- Identifies causes of past events – 1 mark
- Analyzes relationships/patterns in data – 1 mark

5. State any two uses of Time Series Charts. [U] / [CO2]

Any two uses – 1 mark each

- Identifies trends over time – 1 mark
- Forecasts future values – 1 mark

6. Distinguish the objective of descriptive data mining. [U] / [CO1]

- **Summarizes main features of data – 1 mark**
- **Identifies patterns without prediction – 1 mark**

7. Why are data extracts preferred in Tableau? [U] / [CO2]

Any two reasons – 1 mark each

- Improves performance (faster processing) – 1 mark
- Enables offline analysis – 1 mark

8. What is the purpose of Tableau Public? [K] / [CO2]

- **Create and publish visualizations online – 1 mark**
- **Share dashboards publicly for free – 1 mark**

9. Why are map visualizations useful for geographical analysis? [K] / [CO3]

Any two points – 1 mark each

- Displays data based on location – 1 mark
- Helps identify regional patterns – 1 mark

10. Outline the need for using Power BI. [K] / [CO3]

Any two points – 1 mark each

- Provides interactive data visualization – 1 mark
- Supports decision-making with real-time insights – 1 mark

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries 5 marks

11. Explain the workflow and data exploration capability of QlikView. [U] / [CO1]

Workflow – 2 marks

- Data loading from multiple sources – 1 mark
- Data transformation and model creation – 1 mark

Data Exploration – 3 marks

- Associative data model for quick analysis – 1 mark
- Interactive filtering and selection – 1 mark
- Real-time visualization and insights – 1 mark

12. Define Dashboard in Tableau. Explain the key features of a good dashboard.

[K] / [CO2]

Definition – 1 mark

- A dashboard is a collection of visualizations displayed on a single screen

Features – 4 marks

- Clear and simple layout – 1 mark
- Interactive elements (filters/actions) – 1 mark
- Real-time data updates – 1 mark
- Effective use of charts and visuals – 1 mark

13. Explain how dashboards support interactive and visual data interpretation.

[U] / [CO3]

Explanation – 5 marks

- Combines multiple visualizations in one view – 1 mark
- Allows user interaction through filters – 1 mark
- Enables drill-down analysis – 1 mark
- Improves understanding through visual representation – 1 mark
- Supports quick decision-making – 1 mark

14. Describe the role of descriptive analytics in reporting and data interpretation.

[U] / [CO1]

Explanation – 5 marks

- Summarizes historical data – 1 mark
- Identifies patterns and trends – 1 mark
- Supports reporting with visual summaries – 1 mark
- Helps understand business performance – 1 mark
- Provides insights for decision-making – 1 mark

15. Explain the process of resolving data inconsistencies in Power BI and the method used to identify them. [U] / [CO3]

Identification – 2 marks

- Detect missing or duplicate values – 1 mark
- Use data profiling tools in Power Query – 1 mark

Resolution – 3 marks

- Clean and transform data using Power Query – 1 mark

- Remove duplicates and handle null values – 1 mark
- Standardize formats and correct errors – 1 mark

16. Explain how data visualization tools improve the assimilation of business information.

[U] / [CO1]

Explanation – 5 marks

- Converts complex data into visual form – 1 mark
- Makes information easy to understand – 1 mark
- Highlights trends and patterns – 1 mark
- Supports faster decision-making – 1 mark
- Improves communication of insights – 1 mark

17. Differentiate between Symbol Map and Filled Map in Tableau. Explain their features and suitable use cases. [U] / [CO2]

Symbol Map – 2 marks

- Uses symbols to represent data points – 1 mark
- Suitable for showing individual locations – 1 mark

Filled Map – 3 marks

- Uses color shading for regions – 1 mark
- Represents aggregated data by area – 1 mark
- Suitable for regional comparisons – 1 mark

18. Identify the visualization chart best suited for performing time series analysis to show trends. [U] / [CO3]

Chart – 1 mark

- Line Chart

Explanation – 4 marks

- Shows data over continuous time – 1 mark
- Highlights trends clearly – 1 mark
- Useful for forecasting – 1 mark
- Easy to interpret changes over time – 1 mark

[5 × 6 = 30]